

Solution Key to Problem Sheet # 6

Answer.1

Periodic a_k with period $N = 10$

$$a_{11} = a_1 = a_{-1} = 5 \text{ (Since real and even)}$$

$$\sum_{k=-1}^8 |a_k|^2 = 50$$

$$a_1^2 + a_{-1}^2 + a_0^2 + \sum_{k=2}^8 |a_k|^2 = 50$$

$$a_0^2 + \sum_{k=2}^8 |a_k|^2 = 0$$

It means all above coefficients are zero i.e. $a_0, a_2, a_3 \dots \dots a_8 = 0$

$$x(n) = 5e^{\frac{j2\pi n}{10}} + 5e^{-\frac{j2\pi n}{10}}$$

$$x(n) = 10\cos\left(\frac{\pi n}{10}\right)$$

$$A = 10, B = \pi/5$$

Answer.2

$x(n)$ is periodic with period $N = 4$

$$a_k = \frac{1}{4} \sum_{n=0}^3 x(n) e^{-j2\pi nk/4} = \frac{1}{4}$$

$$y(n) = \sum_{k=0}^3 a_k H\left(e^{\frac{j2\pi k}{4}}\right) e^{\frac{j2\pi kn}{4}}$$

$$y(n) = \frac{1}{4} H(e^{0j}) e^{0j} + \frac{1}{4} H(e^{j\pi/2}) e^{\frac{j\pi n}{2}} + \frac{1}{4} H(e^{j\pi}) e^{j\pi n} + \frac{1}{4} H(e^{j3\pi/2}) e^{\frac{j3\pi n}{2}}$$

$$y(n) = \cos\left(\frac{5\pi}{2}n + \frac{\pi}{4}\right)$$

$$y(n) = \cos\left(\frac{\pi}{2}n + \frac{\pi}{4}\right)$$

$$y(n) = \frac{1}{2} e^{j(\frac{\pi}{2}n + \frac{\pi}{4})} + \frac{1}{2} e^{-j(\frac{\pi}{2}n + \frac{\pi}{4})}$$

$$y(n) = \frac{1}{2} e^{j(\frac{\pi}{2}n + \frac{\pi}{4})} + \frac{1}{2} e^{j(\frac{3\pi}{2}n - \frac{\pi}{4})}$$

By comparing

$$H(e^{0j}) = H(e^{j\pi}) = 0$$

$$H(e^{j\pi/2}) = 2 e^{j\pi/4} \quad H(e^{j3\pi/2}) = 2 e^{-j\pi/4}$$

Answer.3

a) $x_1(n) = (-1)^n$ period $N = 2$

$$a_k = \frac{1}{2} (1 - e^{-j\pi k})$$

$$a_0 = 0 \quad a_1 = 1$$

$$y_1(n) = \sum_{k=0}^1 a_k H\left(e^{j\frac{2\pi k}{2}}\right) e^{j\frac{2\pi k n}{2}}$$

$$y_1(n) = 0 + a_1 H(e^{j\pi}) e^{j\pi n}$$

$$y_1(n) = 0 \quad [\text{given } H(e^{j\pi}) = 0]$$

b)

$$x_2(n) = \sin\left(\frac{3\pi}{8}n + \frac{\pi}{4}\right)$$

$$N = 16$$

$$w_0 = \frac{2\pi}{N} = \frac{2\pi}{16} = \frac{\pi}{8}$$

$$x_2(n) = \frac{1}{2j} [e^{j(\frac{3\pi}{8}n + \frac{\pi}{4})} - e^{-j(\frac{3\pi}{8}n + \frac{\pi}{4})}]$$

$$a_3 = \frac{1}{2j} e^{j\pi/4} \quad a_{-3} = \frac{-1}{2j} e^{-j\pi/4}$$

If we take the period from $k=0$ to 15, then coefficients will be only at $k=3$ and 13

$$y_2(n) = \sum_{k=0}^{15} a_k H\left(e^{j\frac{2\pi k}{N}}\right) e^{j\frac{2\pi k n}{N}}$$

$$H\left(e^{j\frac{\pi k}{8}}\right) = 1 \text{ for } k = 3 \text{ and } 13$$

$$y_2(n) = a_3 e^{\frac{j3\pi n}{8}} + a_{13} e^{\frac{-j3\pi n}{8}}$$

$$a_{13} = a_{-3}$$

$$y_2(n) = \frac{1}{2j} e^{j\pi/4} e^{\frac{j3\pi n}{8}} + \left(\frac{-1}{2j}\right) e^{-j\pi/4} e^{\frac{-j3\pi n}{8}}$$

$$y_2(n) = \sin\left(\frac{3\pi}{8}n + \frac{\pi}{4}\right)$$

c)

$x_3(n) = \sum_{k=-\infty}^{\infty} \left(\frac{1}{2}\right)^{n-4k} u(n-4k)$ can be written as

$$x_3(n) = \left[\left(\frac{1}{2}\right)^n u(n) \right] * \sum_{k=-\infty}^{\infty} \delta(n-4k) = g(n) * r(n)$$

$$y_3(n) = g(n) * r(n) * h(n)$$

$$q(n) = r(n) * h(n)$$

We pass $r(n)$ through the filter and convolve the result with $g(n)$.

F.S. coefficient of $r(n)$ are $a_k = \frac{1}{4}$ for all values of k

$$q(n) = \sum_{k=0}^3 a_k H\left(e^{j\frac{2\pi k}{4}}\right) e^{j\frac{2\pi kn}{4}}$$

From given figure, $H\left(e^{j\frac{2\pi k}{4}}\right) = 0$ for all k (0 to 3). So,

$$q(n) = 0$$

$$y_3(n) = g(n) * 0 = 0$$

Answer.4

a) $x(n - n_0)$

Just replace the value of n by $n - n_0$

$$\begin{aligned} a_k &= \frac{1}{N} \sum_{n=\langle N \rangle} x(n) e^{-j\frac{2\pi nk}{N}} \\ a_k &= \frac{1}{N} \sum_{n=0}^{N-1} x(n - n_0) e^{-j\frac{2\pi nk}{N}} \\ &\quad n - n_0 = l \\ a_k &= \frac{1}{N} \sum_{l=-n_0}^{N-1-n_0} x(l) e^{-jk\left(\frac{2\pi}{N}\right)(l+n_0)} \\ &= e^{-jk\left(\frac{2\pi}{N}\right)n_0} \frac{1}{N} \sum_{l=\langle N \rangle} x(l) e^{-jk\left(\frac{2\pi}{N}\right)l} \\ b_k &= e^{-jk\left(\frac{2\pi}{N}\right)n_0} a_k \end{aligned}$$

b) $x(n) - x(n - 1)$

Using delay identity part (a):

$$\begin{aligned} x(n - 1) &\rightarrow e^{-jk\left(\frac{2\pi}{N}\right)} a_k \\ x(n) &\rightarrow a_k \\ b_k &= a_k [1 - e^{-jk\left(\frac{2\pi}{N}\right)}] \end{aligned}$$

c) $x(n) - x(n - \frac{N}{2})$

Multiply the delay in part b with $N/2$

$$b_k = a_k [1 - e^{-j\pi k}]$$

$$b_k = \begin{cases} 2a_k & \text{odd } k \\ 0 & \text{even } k \end{cases}$$

d) $x(n) + x(n + \frac{N}{2})$

Since we are replacing the sign in part c) you can just take the even cases where the values add up.

$$b_k = 2a_{2k}$$

e) $x^*(-n)$

Direct use of properties of Fourier transform.

$$a_k = a_k^*$$

f) $(-1)^n x(n)$

Convert $(-1)^n$ to the form of $e^{2\pi n N/2N}$ since N is even

$$b_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j(\frac{2\pi n}{N})(k - \frac{N}{2})}$$

$$b_k = a_{k - \frac{N}{2}}$$

g) $(-1)^n x(n)$

For N is odd with signal periodic with $2N$

$$b_k = \begin{cases} a_{\frac{k-N}{2}}, & k \text{ is odd} \\ 0, & k \text{ is even} \end{cases}$$

h) $y(n) = \begin{cases} x(n) & n \text{ even} \\ 0 & n \text{ odd} \end{cases}$

We can write $y(n)$ as the following:

$$y(n) = \frac{1}{2} [x(n) + (-1)^n x(n)]$$

For N even we will get

$$b_k = \frac{1}{2} [a_k + a_{\frac{k-N}{2}}]$$

For N odd we get

$$b_k = \begin{cases} \frac{1}{2} [a_k + a_{\frac{k-N}{2}}] & k \text{ odd} \\ \frac{1}{2} a_k & k \text{ even} \end{cases}$$

Answer.5

a) N even $x(n) = -x(n + \frac{N}{2})$

Now the Fourier inverse equation gives us:

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-\frac{j2\pi nk}{N}}$$
$$= \frac{1}{N} \sum_{n=0}^{\frac{N}{2}-1} x(n) e^{-\frac{j2\pi nk}{N}} + \frac{1}{N} \sum_{n=N/2}^{N-1} x(n) e^{-\frac{j2\pi nk}{N}}$$

Replacing n by $2n$ (hence the sum below will be for $n \in \text{Set of Even Numbers}$)

$$= \frac{1}{N} \sum_{n=0}^{N-1} [1 - e^{-j\pi k}] x(n) e^{-\frac{j4\pi nk}{N}} = 0$$

b) $x(n) = -x(n + \frac{N}{4})$

$$a_k = \frac{1}{N} \sum_{n=0}^{\frac{N}{4}-1} x(n) e^{-\frac{j2\pi nk}{N}} + \frac{1}{N} \sum_{n=\frac{N}{4}}^{\frac{N}{2}-1} x(n) e^{-\frac{j2\pi nk}{N}} + \frac{1}{N} \sum_{n=\frac{N}{2}}^{\frac{3N}{4}-1} x(n) e^{-\frac{j2\pi nk}{N}}$$
$$+ \frac{1}{N} \sum_{n=\frac{3N}{4}}^{N-1} x(n) e^{-\frac{j2\pi nk}{N}}$$

$$a_k = \frac{1}{N} \left[\sum_{n=0}^{\frac{N}{4}-1} x(n) + x\left(n + \frac{N}{4}\right) e^{-\frac{j\pi k}{2}} + x\left(n + \frac{2N}{4}\right) e^{-\frac{j2\pi k}{2}} \right. \\ \left. + x\left(n + \frac{3N}{4}\right) e^{-\frac{j3\pi k}{2}} \right] e^{-\frac{j2\pi nk}{N}}$$

$a_k = 0$ For k multiple of 4 like how we get the result for 5 (a).

c) Breaking down a_k in terms of fourier inverse transform and then expanding it as a sum of different exponentials we would get the following:

$$a_k = \frac{1}{N} \left[\sum_{n=0}^{\frac{N}{M}-1} (1 - e^{-j2\pi r} + e^{-j4\pi r} \dots \dots \dots + e^{-j(m-1)\pi r}) \right] e^{-\frac{j2\pi nk}{N}}$$

$$r = k/m$$

$a_k = 0$ For k multiple of m .

Answer.6

a) $x_m(n + Nm) = x_m(n)$

$$\begin{aligned} x_m(n + Nm) &= x\left(\frac{n + mN}{m}\right) \\ &= x\left(\frac{n}{m} + N\right) \\ &= x\left(\frac{n}{m}\right) = x_m(n) \\ n &= 0, \pm m, \pm 2m \end{aligned}$$

$x_m(n)$ is periodic with period mN .

b) $x(n) = v(n) + w(n)$

$$\begin{aligned} x\left(\frac{n}{m}\right) &= v\left(\frac{n}{m}\right) + w\left(\frac{n}{m}\right) \\ x_m(n) &= v_m(n) + w_m(n) \end{aligned}$$

c) $x(n) = e^{-\frac{j2\pi nk_0}{N}}$

$$\begin{aligned} x_m(n) &= e^{-\frac{j2\pi nk_0}{mN}} \\ a_k &= \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-\frac{j2\pi nk_0}{N}} \\ y(n) &= \frac{1}{m} \sum_{l=0}^{m-1} e^{\frac{j2\pi((k_0+lN)n)}{mN}} \\ y(n) &= \frac{1}{m} e^{\frac{j2\pi nk_0}{mN}} \sum_{l=0}^{m-1} e^{\frac{j2\pi nl}{m}} \\ y(n) &= \begin{cases} e^{\frac{j2\pi nk_0}{mN}} & \text{for } n = 0, \pm m, \pm 2m \\ 0 & \text{else} \end{cases} \end{aligned}$$

d) $x_m(n) \rightarrow b_k$

$$b_k = \frac{1}{mN} \sum_{n=0}^{mN-1} x_m(n) e^{-\frac{j2\pi nk}{mN}}$$

Only m^{th} value in above is nonzero.

$$\begin{aligned} b_k &= \frac{1}{mN} \sum_{n=0}^{N-1} x_m(mn) e^{-\frac{j2\pi mnk}{mN}} \\ b_k &= \frac{1}{mN} \sum_{n=0}^{N-1} x(n) e^{-\frac{j2\pi nk}{N}} \\ b_k &= \frac{1}{m} a_k \end{aligned}$$

Answer.7

a) $x(n) \rightarrow a_k$

$$x^*(n) \rightarrow a_{-k}^*$$

$$x(n) \times x^*(n) = |x(n)|^2$$

$$b_k = \sum_{l=\langle N \rangle} a_l a_{l-k}^*$$

b) Yes

Answer 8:

(a) No LTI system exists. The only possible relationship is $y[n] = x[n]^2$. This is a quadratic system (and thus not linear).

(b) The Fourier transform of $x[n]$ is given as: $X(e^{jw}) = \frac{1}{1 - (\frac{1}{2})e^{-jw}}$. Similarly, the

Fourier transform of $y[n]$ is given as: $Y(e^{jw}) = \frac{1}{1 - (\frac{1}{4})e^{-jw}}$. The frequency response of

the LTI system is given by $H(e^{jw}) = \frac{Y(e^{jw})}{X(e^{jw})}$. Thus, the frequency response is

$$H(e^{jw}) = \frac{1 - (\frac{1}{2})e^{-jw}}{1 - (\frac{1}{4})e^{-jw}}, \text{ which is unique.}$$

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(d) The Fourier transform of $x[n]$ is given as: $X(e^{jw}) = e^{j\frac{\pi}{8}} \times 2\pi \sum_{k=-\infty}^{\infty} \delta(w - 2\pi k)$.

Similarly, the Fourier transform of $y[n]$ is given as: $Y(e^{jw}) = 2 \times e^{j\frac{\pi}{8}} \times$

$2\pi \sum_{k=-\infty}^{\infty} \delta(w - 2\pi k)$. The frequency response of the LTI system is given by

$$H(e^{jw}) = \frac{Y(e^{jw})}{X(e^{jw})}. \text{ Thus, the frequency response is } (e^{jw}) = 2, \forall w = 2\pi k, k \in \mathbb{Z}.$$

Thus, the frequency response is not unique.

(e) The Fourier transform of $x[n]$ is given as: $X(e^{jw}) = e^{j\frac{\pi}{8}} \times (2\pi \sum_{k=-\infty}^{\infty} \delta(w - 2\pi k) + \frac{1}{1 + e^{-jw}})$. Similarly, the Fourier transform of $y[n]$ is given as: $Y(e^{jw}) =$

$2 \times e^{j\frac{\pi}{8}} \times (2\pi \sum_{k=-\infty}^{\infty} \delta(w - 2\pi k) + \frac{1}{1 + e^{-jw}})$. The frequency response of the LTI

system is given by $H(e^{jw}) = \frac{Y(e^{jw})}{X(e^{jw})}$. Thus, the frequency response is $(e^{jw}) = 2$.

Thus, the frequency response is unique.

- (f) $x[n] = e^{j\frac{\pi}{2}n}$ and $y[n] = 2(1-j)e^{j\frac{\pi}{2}n}$. The Fourier transform of $x[n]$ is given as:
 $X(e^{jw}) = e^{j\frac{\pi}{8}} \times (2\pi \sum_{k=-\infty}^{\infty} \delta(w - \frac{\pi}{2} - 2\pi k))$. Similarly, the Fourier transform of $y[n]$ is given as: $Y(e^{jw}) = 2 \times (1-j) \times e^{j\frac{\pi}{8}} \times (2\pi \sum_{k=-\infty}^{\infty} \delta(w - \frac{\pi}{2} - 2\pi k))$.
The frequency response of the LTI system is given by $H(e^{jw}) = \frac{Y(e^{jw})}{X(e^{jw})}$. Thus, the frequency response is $H(e^{jw}) = 2(1-j), \forall w = 2\pi k + \frac{\pi}{2}, k \in \mathbb{Z}$. Thus, the frequency response is not unique.
- (g) $x[n] = 0.5 \times (e^{j\frac{\pi}{2}n} + e^{-j\frac{\pi}{2}n})$ and $y[n] = 2(1-j)e^{j\frac{\pi}{2}n}$. The Fourier transform of $x[n]$ is given as: $X(e^{jw}) = e^{j\frac{\pi}{8}} \times (2\pi \sum_{k=-\infty}^{\infty} \delta(w - \frac{\pi}{2} - 2\pi k))$. Similarly, the Fourier transform of $y[n]$ is given as: $Y(e^{jw}) = 2 \times (1-j) \times e^{j\frac{\pi}{8}} \times (2\pi \sum_{k=-\infty}^{\infty} \delta(w - \frac{\pi}{2} - 2\pi k))$. The frequency response of the LTI system is given by $H(e^{jw}) = \frac{Y(e^{jw})}{X(e^{jw})}$. Thus, the frequency response is $H(e^{jw}) = 2(1-j), \forall w = 2\pi k + \frac{\pi}{2}, k \in \mathbb{Z}$. Thus, the frequency response is not unique.
- (h) The relationship between $y[n]$ and $x[n]$ is: $y[n] = x[n-3] + x[n+3]$. The frequency response of the LTI system is given by $H(e^{jw}) = \frac{Y(e^{jw})}{X(e^{jw})}$. We have, $Y(e^{jw}) = X(e^{jw})(e^{-3jw} + e^{3jw})$. Thus, the frequency response is $H(e^{jw}) = e^{-3jw} + e^{3jw}$, which is unique.
- (i) There is not LTI relationship between $x[n]$ and $y[n]$. The system is not LTI.